UNITED INTERNATIONAL UNIVERSITY

Department of Computer Science and Engineering (CSE)

Course Syllabus

Part A: Introduction		
1	**Course Title**	Web Programming
2	**Course Code**	CSE 4165
3	**Pre-requisites**	CSE 1115, CSE 1116
4	**Course Type**	Mandatory Course
5	**Credit Hours**	3.00
6	**Contact Hours**	2.5 Hours/Week
7	**Semester**	6th
8	**Total Marks**	100
9	**Course Instructor's Information**	Mahmudul Hasan mahmudul@cse.uiu.ac.bd Room: 836(A)
10	**Course Rationale**	The world of software development has shifted vastly to web applications & cloud based apps. Nowadays software is mainly developed on client server based architecture. Even evolving use of mobile applications also added new paradigm to this concept. Currently without proper concept & expertise in web communication, it is not possible to flourish as a Software Engineer. This course will enhance concept & expertise of the students in this crucial development paradigm.
11	**Course Objectives**	The course is designed to provide the background of the following topics 1. Provide the basic concepts of web programming 2. Develop insight & expertise on web application 3. Design a web application architecture 4. Able to develop a real time web application using all the components
Part B: Content of the Course		

12	**Course Contents (approved by UGC)**	This course will introduce students with client server architecture & the basic of web communication. They will learn in detailed about different components of a web application. For client side programming they will go through HTML, CSS & JavaScript. And for server side programming & database communication they will work with Python and Python framework as server-side technology, integrated with a MySQL database. They will learn to operate with the complete web application eco-system. Students will learn to design a real time application using these components. Along with development, students will learn techniques for webpage optimization to improve loading speed and performance. Finally, students will be introduced to website hosting strategies and cPanel, and will demonstrate a complete project as the final outcome of the course.
13	**Course**	

	Outcomes (COs)	**COs**	**Description**
		CO1	**Learn** the basics of web development.
		CO2	**Apply** modern tools and techniques for web development.
		CO3	**Develop** a modern web based systems.

14 Mapping of COs and Program outcomes

COs	**Program Outcomes(POs)**											
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1				C								
CO2					C							
CO3							C					

15 Mapping COs with Teaching-Learning and Assessment Strategy

Class	**Topics/Assignments**	**COs**	**Teaching-Learning Strategies**	**Assessment Strategies**
Class 1	HTML: structure, tags, headings, links, images, lists	CO1	Lecture/Group Discussion/ Exercise	Class Tests/Assignments/Quizzes /Exam
Class 2	HTML: forms, inputs, tables, semantic tags, media	CO1	Lecture/Group Discussion/ Exercise	Class Tests/Assignments/Quizzes /Exam
Class 3	CSS: selectors, box model, colors, fonts, units	CO1	Lecture/Group Discussion/ Exercise	Class Tests/Assignments/Quizzes /Exam

Class 4	CSS: Flexbox, Grid, responsive design, Bootstrap	CO2	Lecture/Group Discussion/ Exercise	Class Tests/Assignments/Quizzes /Exam
Class 5	JavaScript: variables, data types, functions, arrays, strings	CO2	Lecture/Group Discussion/ Exercise	Class Tests/Assignments/Quizzes /Exam
Class 6	JavaScript: DOM, events, objects, JSON	CO2	Lecture/Group Discussion/ Exercise	Class Tests/Assignments/Quizzes /Exam
MIDTERM EXAMINATION				
Class 7	Python Basics	CO3	Lecture/Group Discussion/ Exercise	Class Tests/Assignments/Quizzes /Exam
Class 8	Python Class, Library, tupple, dictionary	CO3	Lecture/Group Discussion/ Exercise	Class Tests/Assignments/Quizzes /Exam
Class 9	Django basic, databases intergration, SQL and ORMs	CO3	Lecture/Group Discussion/ Exercise	Class Tests/Assignments/Quizzes /Exam
Class 10	Django forms, validation and CRUD, CSRF protection and form integration	CO3	Lecture/Group Discussion/ Exercise	Class Tests/Assignments/Quizzes /Exam
Class 11	Django permissions and deployment	CO3	Lecture/Group Discussion/ Exercise	Class Tests/Assignments/Quizzes /Exam
Class 12	Project Evaluation, Term Final discussion	CO3	Lecture/Group Discussion/ Exercise	Class Tests/Assignments/Quizzes /Exam
FINAL EXAMINATION				

Part C: Assessment and Evaluation Methods

Assessment Strategy	**Assessment Types**	**Marks**
Formative Assessment	Attendance	5%
	Project with viva	20%
Summative Assessment	Quiz + homework	15%
	Mid Term	30%
	Final Exam	30%

Grading System

Letter Grade	Marks %	Grade Point	Letter Grade	Marks%	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Part D: Learning Resources

Text Book	Learning PHP, MySQL & JavaScript, 5th Edition (May, 2018) by Robin Nixon
Reference	Laravel: Up & Running, 3rd Edition (2023) by Matt Stauffer
LMS URL	

Appendix-1: Program outcomes

POs	Program Outcomes
PO1	An ability to apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PO2	Identify, formulate, research and analyze complex engineering problems and reach substantiated conclusions using the principles of mathematics, the natural sciences and the engineering sciences.
PO3	An ability to design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety and of cultural, societal and environmental concerns.
PO4	An ability to conduct investigations of complex problems, considering experimental design, data analysis and interpretation and information synthesis to provide valid conclusions.
PO5	An ability to create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling, to complex engineering activities with an understanding of their limitations
PO6	An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO7	An ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.
PO8	An ability to apply ethical principles and commit to the professional ethics, responsibilities and the norms of the engineering practice.
PO9	An ability to function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings.
PO10	An ability to communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.
PO11	An ability to demonstrate knowledge and understanding of engineering and management principles and apply these to one's work as a team member or a leader to manage projects in multidisciplinary environments.
PO12	An ability to recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.